

Simultaneous Equations & Inequalities



Linear simultaneous equations

- 1 Solve the system: $3x + 2y = 16$ and $x - y = 2$.
 - 2 Solve the system: $4x - 3y = 1$ and $2x + y = 9$.
 - 3 Solve the system: $3x + 4y = 18$ and $x - 2y = -4$.
 - 4 For the system $x + 3y = 11$ and $2x - y = 1$:
 - a Solve for x and y .
 - b State whether the point of intersection lies in the first quadrant.
-

Solving a linear and a quadratic equation simultaneously

- 5 Solve the system: $y = x + 1$ and $y = x^2 - 5$.
 - 6 Solve the system: $2x - y = 1$ and $x^2 + y^2 = 29$.
 - 7 Solve the system: $y = 2x^2$ and $y = x + 1$.
-

Using the discriminant to determine intersections

- 8 Find the values of k for which the line $y = kx$ is tangent to the curve $y = x^2 + 4$ (i.e. meets it at exactly one point).
 - 9 Find the values of c for which the line $y = 3x + c$ does not intersect the curve $y = x^2 + 2x + 5$.
 - 10 Find the values of m for which the line $y = mx + 2$ intersects the curve $y = x^2 + 3x + 7$ at two distinct points.
-

Solving linear inequalities

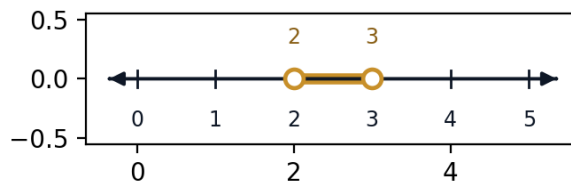
- 11 Solve $5x - 3 > 2x + 9$.
- 12 Solve $-3x + 7 \leq 1$, being careful with the direction of the inequality.

13 Solve $2(x - 3) + 5 \geq 3x - 4$.

14 Solve $4 - 2x < 10$.

Solving quadratic inequalities

15 Solve $x^2 - 5x + 6 < 0$, and represent the solution on a number line.



16 Solve $x^2 \geq 9$.

17 Solve $2x^2 - 3x - 2 \leq 0$.

18 For the inequality $x^2 - 4x - 5 \geq 0$:

- a Factor the quadratic and find its roots.
- b State the solution set for the inequality.

Combining a quadratic inequality with a linear constraint

19 Solve the compound inequality $x^2 - 2x - 8 \leq 0$ and $x > -1$ simultaneously, giving the combined solution set.

20 Solve the compound inequality $x^2 - 7x + 10 \leq 0$ and $x < 4$ simultaneously, giving the combined solution set.

21 Solve the compound inequality $x^2 - 9 \leq 0$ and $x \geq -2$ simultaneously, giving the combined solution set.

Simultaneous equations from a real-world constraint

22 A rectangle has perimeter 34 and diagonal length 13. Find its length and width.

23 Two numbers have a sum of 20 and a product of 91. Find the two numbers.

24 Two positive numbers have a difference of 5 and a product of 84. Find the two numbers.

A well-designed word problem: break-even analysis

- 25** This question has two parts. A company's costs are modeled by $C(x) = 500 + 8x$ dollars and revenue by $R(x) = 20x - 0.02x^2$ dollars, where x is the number of units sold. (a) Set up and solve the inequality $R(x) > C(x)$ to find the range of production levels for which the company makes a profit. (b) Explain, in terms of the model, why profit is not simply increasing forever as x grows.